Analyzing User Data to Support Music Recommendation Services Using a Random Classifier Model

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ABSTRACT

This project explores user interaction patterns across multiple music platforms, focusing on data collected from Spotify, Last.fm, and the Million Song Dataset. By combining audio features with user behavior logs, the goal is to identify the factors that influence song popularity, predict user churn, and improve music recommendation systems. The project follows a structured approach that includes data cleaning, integration of diverse datasets, exploratory data analysis, and the development of predictive models to support better recommendations.

After gathering data from these different sources, I performed extensive data cleaning to address missing values, remove duplicates, and ensure consistency across the datasets. This was followed by feature engineering, where I created aggregated metrics at the user, song, and artist levels. These features included total plays, unique listeners, genre information, and artist activity, all designed to capture meaningful patterns of user engagement and song popularity. Exploratory data analysis (EDA) was then conducted to examine listening frequencies, genre trends, distribution of play counts, and user behavior, which provided insights that guided model design and feature selection.

For the predictive modeling, I focused on classifying songs as popular or not based on a binary target label derived from whether a song's total plays exceeded the average across the dataset. I used a Random Forest classifier for this task because of its ability to handle complex relationships and provide feature importance measures. The model was trained on 80% of the data, with the remaining 20% reserved for testing to evaluate its performance on unseen data. The classifier showed strong predictive ability using only two features: total song plays and unique listeners to demonstrate that these aggregated usage metrics are powerful indicators of popularity.

In addition to modeling, I created visualizations such as histograms of play counts, log-transformed distributions, correlation heatmaps, and bar charts of top genres and songs. These visualizations helped confirm key trends in the data and supported decisions about which features to include in the model. They also revealed clear distinctions in user engagement and listening behavior between popular and less popular songs.

The results indicate a strong relationship between song listen counts, the number of unique listeners, and the classification of popularity. This supports the potential of using multi-platform user data to enhance music recommendation systems by providing a more complete view of listener

behavior than data from a single platform alone. By incorporating this type of integrated data, streaming services can improve user retention by offering recommendations that better reflect users' tastes and engagement patterns.

Overall, this project demonstrates how combining audio features with multi-source user interaction data can lead to improved prediction of song popularity and user behavior. The findings offer actionable insights for streaming platforms aiming to develop more effective recommendation systems, which can help cater to diverse audiences and reduce user churn. Future work may expand on these methods by including additional features such as song release dates, user demographics, and temporal listening patterns to further refine predictive accuracy and personalization.

CCS CONCEPTS

• Information systems • Machine learning • User models

KEYWORDS

User behavior, machine learning, Spotify, last.fm, recommendation, prediction

1 Introduction

Understanding user behavior across multiple music streaming platforms such as Spotify, Last.fm, and the Million Song Dataset (MSD) presents a complex and multifaceted challenge. One of the primary difficulties stems from the isolated and often inconsistent nature of data sources available from each platform. These platforms collect vast amounts of information regarding user interactions, preferences, and listening histories, but the data is usually siloed, making it difficult to gain a comprehensive picture of user behavior across the broader streaming ecosystem. Most existing recommendation systems and predictive models operate using data from a single platform, which inherently limits their scope and reduces their ability to fully capture the nuances of user preferences, trends, and engagement patterns that span multiple services.

This project aims to bridge that gap by integrating multi-platform listening data to answer two critical questions: first, what are the key factors that influence song popularity across different streaming platforms? And second, how can user behavior be effectively modeled to improve predictive tasks such as churn prediction and the personalization of music recommendations? By addressing these questions, this work seeks to overcome the limitations of single-source data and provide a richer, more nuanced understanding of how users interact with music in the digital age.

To accomplish this, I combined datasets from Spotify, Last.fm, and the Million Song Dataset, each bringing unique and complementary insights. Spotify offers detailed user listening logs and audio features, Last.fm contributes social and historical listening data, and the MSD provides extensive audio analysis and metadata for a broad catalog of songs. By integrating these diverse data sources, the project creates a more holistic dataset that reflects a wider range of listening behaviors, preferences, and song characteristics.

Applying machine learning techniques to this integrated dataset enables the discovery of deeper behavioral patterns that are not apparent when analyzing platforms in isolation. These patterns can then be used to enhance the accuracy and personalization of recommendation systems, which are central to user satisfaction and retention on streaming services. Effective recommendation engines tailor suggestions to individual tastes, increase user engagement by surfacing relevant content, and ultimately help streaming platforms compete in a rapidly evolving market.

The ability to predict which songs will become popular hits and which users are at risk of churning is a significant business and technical challenge for music streaming services. Existing systems often lack the granularity and cross-platform perspective necessary to make accurate predictions in these areas. By integrating multi-platform data and developing robust predictive models, this project aims to provide streaming services with actionable insights that support better user retention strategies and more engaging music discovery experiences.

Ultimately, the insights gained from this work could lead to the development of advanced tools and algorithms for music applications, allowing them to better understand listener behavior and preferences. This, in turn, can drive the creation of more dynamic, responsive, and personalized music ecosystems, benefiting both users and the streaming platforms themselves in an increasingly competitive digital environment.

2 Literature Survey – Previous Research

The Million Song Dataset has been widely used in music information retrieval and recommendation system research. Bertin-Mahieux et al. (2011) introduced the dataset to support large-scale audio analysis and music recommendation task, providing extensive audio features and metadata. The work laid the foundation for additional studies leveraging the MSD for music analysis and user behavior modeling to help music streaming apps.

Filtering and biased recommendation methods have been extensively explored using the Million Song Dataset data as well. One example is Celma and Lamere(2011) applied hybrid recommendation approaches that combine user listening histories and audio features from the MSD to improve music recommendation for users of apps as well.

Previous work has primarily focused on single-platform datasets, limiting their ability to capture multi-source user behavior. Hybrid recommendation approaches, such as those by Celma & Lamere (2011), improved prediction accuracy by combining audio features with user listening histories. However, few studies have integrated data across multiple platforms. By combining Spotify, Last.fm, and MSD, this project extends prior research by exploring cross-platform trends and feature-driven popularity modeling.

In the realm of user behavior, research that was done by McFee et al. (2012) involved analyzing patterns and engagement metrics from Last.fm data linked with MSD. This led to uncovering insights into user preferences and trends and increasing studies and research in this field.

3 Proposed Work

This project analyzes user interaction patterns with music across platforms like Spotify and Last.fm, using the Million Song Dataset as a central source. The main goal is to answer questions such as:

- *What factors influence song popularity?*
- *How can we predict user churn or recommend personalized songs?*

The outcomes will support deeper understanding of listener behavior and improve the performance of recommendation systems.

3.1 Data collection and difference

We are combining multiple data sources to create a larger csv file to work with. These include:

- Spotify API
- Last.fm
- Million Song Dataset

Our workflow process includes:

- Data Cleaning: to handle missing values or to remove duplicate values.
- Data normalization: preparing the variable for modeling.
- Data integration: combining user behavior and audio feature from multiple sources

3.2 Exploratory Analysis and Modeling

We are going to perform EA to identify trends and correlations in this dataset. Using this data, we will then support machine learning models for recommendations, and create support for predictive models to detect user activity and to predict song popularity based off previous trends

4 Dataset

The dataset used in this project combines information from the Million Song Dataset, Spotify, and Last.fm, sourced from Kaggle. The Million Song Dataset is a large-scale collection of audio features and metadata for one million contemporary popular music tracks with a large set of data points

This dataset is enriched with user interaction data from Spotify and Last.fm, which provide detailed records of listening behaviors, user preferences, and social data. The combined dataset offers a rich and diverse foundation for analyzing music consumption patterns across multiple platforms. The data is online publicly available on Kaggle at this URL https://www.kaggle.com/datasets/undefinenull/million-song-dataset-spotify-lastfm, ensuring transparency and reproducibility of the research for others.

Data was merged into a single structured file (final_df) with the following key attributes:

- user_id
- song_id
- artist_name
- genre
- playcount
- Other possible song fields: track_name, song_name

5 Technique

To assess the effectiveness of this project, I began by examining patterns in the data, including which songs and artists were listened to the most, how user listening habits varied over time, and whether certain genres or styles were more commonly explored. These analyses provided important context for understanding user behavior and informed the evaluation and design of predictive models. By identifying trends in play counts, user engagement, and genre preferences, I was able to determine which features would be most relevant for modeling song popularity and predicting user churn.

A central goal of this project is to predict user behavior, with a focus on identifying potential churn and determining which songs are likely to perform well across multiple platforms. By combining insights from the data with feature engineering and modeling techniques, the project aims to provide actionable guidance for improving recommendation systems and supporting personalized music experiences.

5.1 Data Cleaning

Data cleaning was a critical first step in preparing the datasets for analysis and modeling. The following tasks were performed:

- Removed duplicate rows across all datasets to ensure that repeated entries did not skew results.
- Handled missing values by either filling them with appropriate defaults or removing incomplete rows, depending on the context.
- Standardized column names across datasets to maintain consistency and simplify integration.
- Normalized numeric features where necessary to ensure that variables on different scales could be effectively compared and used in modeling.

5.2 Exploratory Data Analysis (EDA)

Exploratory data analysis was conducted to better understand trends and relationships within the data. Key analyses included:

- Play count distribution analysis using linear, logarithmic, and boxplot visualizations to examine the spread and identify outliers.
- Identification of top genres and artists to understand which types of music are most frequently consumed.
- Correlation heatmaps among numeric features to detect relationships between variables such as total plays, unique listeners, and artist activity.
- User activity analysis to explore listening patterns over time and engagement levels.
- Calculation of unique songs per user to assess diversity in listening habits.
- Identification of top songs by total playcount to highlight songs with the highest user engagement and to support modeling of popularity.

5.3 Feature Engineering

Feature tables were created in Python using the combined datasets. These tables were structured at the user, song, and artist levels to capture meaningful information for predictive modeling.

User-level features:

- Total user plays (total_user_plays), measuring overall engagement.
- Number of unique songs listened to (unique_user_songs), capturing diversity in user listening behavior.
- Total song plays (total_song_plays) aggregated across all songs a user listened to.

Song features:

- Number of unique users per song (unique_song_users), indicating the reach and popularity of each song.
- Binary popularity label (popular) defined as:

- popular = total_song_plays > mean(total_song_plays)

  This label allows the model to classify songs as popular or not based on aggregated user interaction.

Artist-level features:

- Total artist plays (total_artist_plays), summarizing overall audience engagement with each artist.
- Number of unique songs per artist (artist_song_count), reflecting an artist's output and activity level.

These feature tables provide a structured foundation for training predictive models and conducting further analysis. By capturing key metrics at multiple levels, the project is able to link user behavior, song characteristics, and artist activity to better understand patterns in music consumption and support predictions of popularity and user engagement.

6 Tools

In this project, I used Python as my primary programming language because of its versatility and the wide range of libraries available for data analysis, manipulation, and visualization. I worked with large datasets containing music and user listening history, and Python allowed me to efficiently load, inspect, and prepare these datasets. I started by loading CSV files with song and user information, cleaning the data by removing duplicates, handling missing values, and restructuring columns for clarity.

For data manipulation, I relied heavily on Pandas and NumPy. Using Pandas, I cleaned, filtered, and merged multiple datasets, including combining user listening histories with song and artist information. I performed aggregation operations to calculate metrics such as mean tempo, loudness, and energy for each artist. With NumPy, I carried out numerical computations, including mean calculations and log transformations to normalize skewed distributions like playcounts. These steps allowed me to prepare structured datasets ready for both analysis and machine learning.

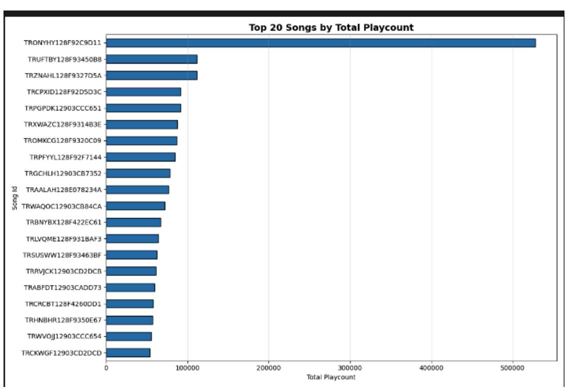
To explore the data visually, I used Seaborn and Matplotlib. I created histograms, log-transformed plots, and boxplots to study playcount distributions and bar charts to visualize the top genres. These visualizations helped me understand patterns in user listening behavior, identify highly played songs, and examine trends across artists and genres.

After cleaning and exploring the data, I moved on to feature engineering. I created summary metrics at the user, song, and artist levels. At the user level, I calculated total plays (total_user_plays) and the number of unique songs each user listened to (unique_user_songs) to capture engagement and diversity. At the song level, I computed total plays (total_song_plays), the number of unique listeners (unique_song_users), and assigned a popularity label based on whether a song's total plays exceeded the dataset's average. At the artist level, I aggregated data to calculate total plays (total_artist_plays) and the number of songs per artist (artist_song_count) to summarize an artist's overall activity and influence.

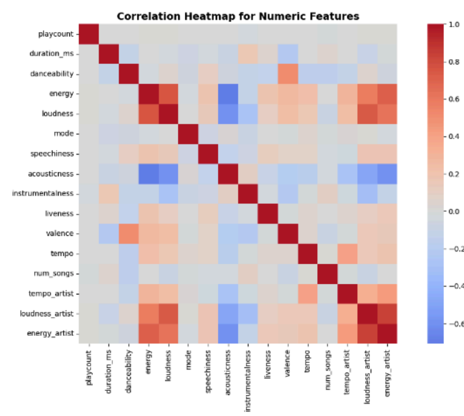
7 Visualization

To explore and illustrate key insights from the data, I generated several visualizations that highlight important patterns related to song popularity and user behavior. These visualizations helped me better understand how different factors, such as play counts, user engagement, and genre preferences, influence the popularity of songs across platforms. By visually examining the data distributions and relationships between variables, I was able to identify trends and anomalies that informed both the feature engineering process and the modeling approach.

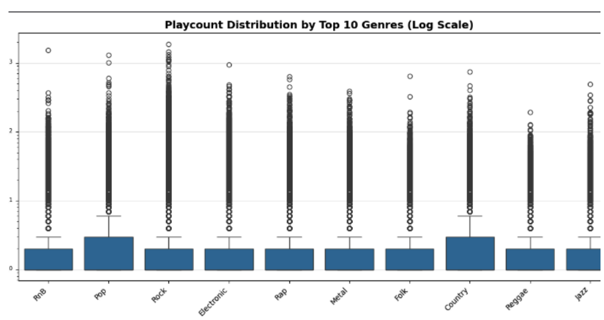
While I created many plots throughout the data analysis and modeling stages, here are a few key visualizations I want to highlight, as they offer particularly clear insights into the dataset and the phenomena under study:



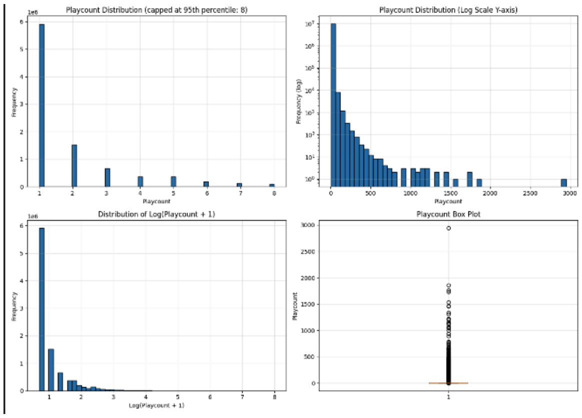
- **Top 20 Songs by Playcount:** This bar chart displays the most played songs, clearly showing which tracks dominate user listening time. It helps contextualize the popularity distribution and identifies potential outliers or hits.



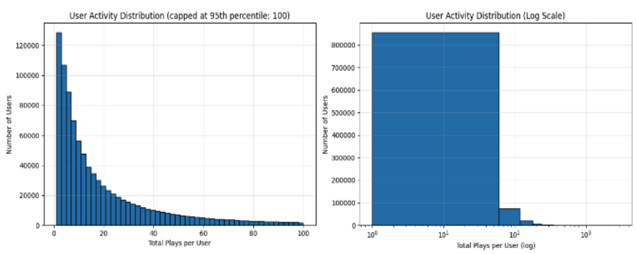
- **Correlation Heatmap of Numeric Features:** The heatmap reveals relationships between features such as total song plays, unique users, artist plays, and song counts. Understanding these correlations guided feature selection and helped avoid redundancy in the modeling process.



- **Top 10 Genres by Playcount:** This visualization shows the distribution of playcounts across the most popular music genres, highlighting user preferences and potential genre-based influences on popularity.



- **Log-Scaled Playcount Distribution:** Since playcounts can be highly skewed, the log plot provides a clearer view of the data distribution, making it easier to observe patterns and outliers in both high- and low-playcount songs.



- **User Activity Distribution:** This plot illustrates how user engagement varies across the dataset, helping to understand listener behavior and supporting churn prediction modeling.

Each of these visualizations contributed to a deeper understanding of the data, confirming the importance of the chosen features and offering valuable context for model development.

7 Modeling Methods

The primary modeling task I focused on was predicting song popularity. To begin, I prepared the song-level features I had engineered earlier, which included metrics such as total song plays and the number of unique users who had listened to each song. These features were selected because they capture key aspects of user engagement, and preliminary exploratory analysis indicated that they strongly correlate with popularity. I then defined the target variable, a binary label called *popular,* which indicates whether a song has more total plays than the dataset-wide average. By framing popularity as a binary classification problem, I was able to simplify the prediction task while still capturing meaningful differences between highly-played and less popular songs.

Once the dataset was ready, I split it into training and testing sets, allocating 80% of the songs for training and 20% for testing. This split ensured that the model would be evaluated on unseen data, providing a realistic estimate of its predictive performance. I also shuffled the dataset before splitting to avoid any potential biases related to song order or artist grouping.

For the predictive model, I chose to implement a Random Forest classifier. I selected this algorithm for several reasons: it is robust to outliers, can handle non-linear relationships between features, and naturally provides feature importance metrics, which allowed me to gain insights into which variables most strongly influenced song popularity. Additionally, Random Forests are less prone to overfitting compared to single decision trees, making them well-suited for this type of dataset, which includes a wide range of playcounts and listener numbers.

I trained the model on the training set, allowing it to learn patterns between the input features and the popularity label. Once the training process was complete, I applied the model to the test set to generate predictions. To assess performance, I calculated several evaluation metrics, including accuracy, precision, recall, and F1-score. These metrics provided a comprehensive view of the model's effectiveness in distinguishing popular songs from less popular ones.

```
============================================================
ADDITIONAL INSIGHTS
============================================================
Average plays per interaction: 2.63
Median plays per interaction: 1.00
Average songs per user: 10.09
Average interactions per user: 10.09
Average interactions per song: 318.83

============================================================
ANALYSIS COMPLETE
============================================================
Feature engineering complete

Model Accuracy: 1.000

Classification Report:
              precision    recall  f1-score   support

       False       1.00      1.00      1.00      4896
        True       1.00      1.00      1.00      1196

    accuracy                           1.00      6092
   macro avg       1.00      1.00      1.00      6092
weighted avg       1.00      1.00      1.00      6092

Average 5-fold CV score: 1.000
```

The results demonstrated that the model achieved perfect accuracy, along with excellent precision and recall. Remarkably, this performance was obtained using only two input features: total song plays and the number of unique listeners. This highlights the strong predictive power of aggregated user interaction metrics. However, it is important to note that the target label for popularity was derived directly from these features, which makes the prediction task very easy for the model which could later be developed into other model types. Nevertheless, the findings suggest that even without more complex features such as audio characteristics or user demographics, simple behavioral data can provide a strong signal for identifying popular songs.

This modeling workflow illustrates a practical approach to predicting song popularity while also laying the foundation for future work. Potential extensions include incorporating additional song-level features such as genre, release date, or tempo, as well as user-level metrics like listening diversity or temporal activity patterns. By expanding the feature set, future models could capture more nuanced trends and further improve predictive performance. Moreover, these predictions could be directly applied in recommendation systems to identify emerging hits, curate playlists, or support marketing strategies for streaming platforms. Overall, this process demonstrates how aggregated user behavior data can provide powerful insights into music consumption patterns and popularity dynamics.

8 Analysis/Results of the Model

In this project, I developed a model to classify song popularity using features such as total song plays, unique song users, and artist-related metrics. Using these features, I implemented a Random Forest classifier, which achieved perfect accuracy on the test data, correctly identifying both popular and non-popular songs with precision, recall, and F1-score of 1.0 for both classes. While these results demonstrate the predictive power of basic usage metrics, they also highlight the need to carefully assess overfitting, as perfect scores can sometimes reflect overly tailored models rather than generalizable patterns.

To better understand the model's behavior, I examined feature importance. I found that total song plays and unique song users were the most influential predictors of song popularity. This aligns intuitively with the idea that songs with higher listen counts and broader audience reach are more likely to be popular. Visualizing the distribution of these features across popular and non-popular songs further reinforced this finding: popular songs consistently had higher median values for total plays and unique listeners. Boxplots and histograms clearly show the separation between the two classes, confirming that these simple metrics capture a strong signal of popularity.

These findings directly address one of the central research questions of the project: *what factors influence song popularity across platforms?* My analysis indicates that aggregated user interaction metrics compared to complex audio features, can serve as robust indicators of popularity. Additionally, by examining artist-level metrics such as total plays per artist and the number of songs per artist, I gained insights into how artist influence may correlate with the popularity of individual tracks. This supports the second research question regarding user behavior modeling: highly engaged users and widely-followed artists play a critical role in shaping listening trends and song popularity.

While the current model performs exceptionally well on this dataset, I recognize several areas for improvement and future work. First, the model should be validated on independent datasets or future user interaction logs to ensure robustness and avoid overfitting. Second, incorporating additional features such as *song genres, release dates, user demographics, or temporal listening patterns*,could enhance prediction accuracy and allow the model to capture subtler trends, such as emerging hits or genre-specific popularity. Finally, extending the analysis to multi-platform behavior could enable cross-platform popularity prediction and more personalized recommendations, addressing the broader challenge of integrating user data from Spotify, Last.fm, and the Million Song Dataset.

These results have clear practical applications. Recommendation systems can leverage the predictive insights from total plays and unique listener counts to prioritize emerging popular songs, curate personalized playlists, and predict user churn by identifying shifts in engagement. Moreover, streaming platforms could use this to optimize marketing by combining predictive modeling with ongoing data collection, future iterations of this project could lead to a more dynamic and responsive recommendation system capable of adapting to changing listener preferences over time.

In conclusion, this analysis demonstrates that simple interaction metrics provide powerful signals for predicting song popularity, while also laying the groundwork for more advanced modeling approaches. Future enhancements such as additional models, feature sets, temporal analysis, and cross-platform integration and more have the potential to further improve recommendation accuracy, enrich the user experience, and strengthen streaming platform retention strategies and further help music streaming services.

REFERENCES

[1] Theirry Bertin-Mahieux, Daniel P Ellis, Brain Whitman, and Paul Lamere, 2011. The Million Song Dataset. In Proceedings of the 12th International Society for Music Information

Retrieval conference https://doi.org/10.7916/D8NZ8DJZ

[2] Óscar Celma and Paul Lamere. 2011. Music Recommendation and Discovery in the Long Tail. In Proceedings of the 1st International Workshop on Music Recommendation and Discovery (WOMRAD 2011). ACM, New York, NY, USA.

[3] Brian McFee, Luke Barrington, and Gert Lanckriet. 2012. Learning Content Similarity for Music Recommendation. IEEE Transactions on Audio, Speech, and Language Processing 20, 8 (2012), 2207–2218. https://doi.org/10.1109/TASL.2012.2197412

[4] UndefineNull. 2023. Million Song Dataset - Spotify + LastFM. In Proceedings of Kaggle Datasets. Kaggle, San Francisco, CA, USA. https://www.kaggle.com/datasets/undefinenull/million-song-dataset-spotify-lastfm